

Designing for Empathy in VR: Navigating the Temporal Tension Between State and Trait-Oriented Approaches

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Abstract

Virtual Reality (VR) has been widely regarded as a promising medium for fostering empathy due to its immersive and interactive capabilities in simulating lived experiences. However, many efforts to design effective empathic VR experiences have encountered persistent challenges, largely stemming from a lack of attention to the temporal dynamics of empathy—particularly the distinction between transient state empathy and more enduring trait empathy. Empathy unfolds across multiple stages, each characterized by distinct features and requiring different design and measurement strategies. Despite its significance, this temporal progression remains insufficiently addressed in the literature, leading to conceptual and methodological inconsistencies across studies. To address this gap, the present study synthesizes existing insights on the temporal structure of empathy, outlines how different components evolve over time, and offers design and evaluation strategies tailored to each stage. In doing so, it provides a clearer foundation for developing temporally coherent and theoretically grounded empathic VR interventions.

CCS Concepts

• **Human-centered computing** → **Collaborative and social computing theory, concepts and paradigms; HCI theory, concepts and models; Virtual reality.**

Keywords

Virtual Reality; Empathic VR; User-Centered Design; Trait Empathy; State Empathy

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1 Introduction

Empathy is a complex mental phenomenon that bridges one's own perspectives and experiences with those of others [6]. It engages multiple pathways—emotional resonance, contextual understanding, embodied response, and prosocial motivation—that are particularly vital in an increasingly diverse and interconnected world [18]. Given the need to activate these pathways in a context-rich and experiential manner, virtual reality (VR) stands out among various approaches for its exceptional ability to simulate real-world situations with immersive fidelity [39].

Despite promising early evidence[39], not all recent efforts to enhance empathy through VR have proven successful, as disagreements have emerged regarding outcomes, design considerations, and even the fundamental capacity of VR to serve as an empathic instrument [37][23]. These inconsistencies point to the need for a critical re-examination of the conceptual assumptions and methodological approaches underlying empathic VR research.

Existing literature suggests the inconsistencies may stem from a common oversight: a failure to recognize that “empathy” refers to temporally distinct constructs—ranging from transient, context-dependent reactions to stable dispositional traits—each requiring different intervention and measurement strategies [6]. Empathy is mediated by at least five distinct neural pathways—encompassing emotional, cognitive, and somatic processes—that do not necessarily unfold or consolidate on the same timescale[18][23]. This complexity is further compounded by insufficient differentiation between trait empathy, a stable dispositional tendency, and state empathy, a transient, context-elicited reaction. This conflation has led to a common methodological mismatch: many studies aiming to elicit state empathy through short-term VR exposure have relied on instruments originally designed to assess trait empathy. Unsurprisingly, this misalignment often results in null findings [17].

While these conceptual distinctions help explain the challenges of designing effective empathic VR experiences, crucial temporal questions remain underexplored. Specifically, it is unclear how trait and state empathy differ in their temporal trajectories, or how



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the neural mechanisms underlying emotional, cognitive, and somatic empathy consolidate over time. Without such clarity, both the design and evaluation of empathic VR experiences remain vulnerable to misalignment. Addressing this gap, the present study examines (1) how different dimensions of empathy evolve over time, (2) how to measure these dimensions effectively with temporal considerations in mind, and (3) how to design VR scenarios for these dimensions of empathy.

2 Empathy, VR, and Time

2.1 Empathy and Temporal Dynamics

The complexity in defining empathy is inherent in its conceptual history. Originating in aesthetic practice as the notion of “feeling into” a work of art [25], the term later evolved to describe the cognitive effort to “put oneself in another’s position” [21], and was at times associated with motor mimicry and imitation [7]. Across these shifts, empathy has been used to describe various phenomena that share a central feature: directed affective, cognitive, or somatic engagement with the internal states of others. Contemporary neuroscience affirms this multiplicity. Brain studies show that empathy is mediated by multiple distinct pathways—commonly including affective, cognitive, somatic, compassionate, and emotion regulation mechanisms [18]. Somatic empathy often functions as an automatic, immediate response via mirror neural mechanisms upon witnessing others’ emotions or suffering [33][3]. Affective empathy follows with emotional resonance, while cognitive and compassionate empathy take longer to consolidate, as they depend on contextual reasoning and value-based reflection [23]. Emotion regulation operates at a meta-level, modulating the entire empathic process.

Moreover, when examined across longer time spans, the temporal dynamics of empathy introduce an additional layer of complexity. As early as the mid-20th century, scholars recognized that the term empathy can refer to both skill and capacity, process and outcome, as well as the stable trait and the transient state [27][6][1]. The state of empathy functions as a situation-specific experience of understanding and sharing another person’s emotions, and it can change significantly following a single exposure to an immersive experience. Over time, the affective component of state empathy tends to diminish, whereas the cognitive connection—often internalized as an “in-group” perspective—may persist, fostering a sustained inclination to empathize with that particular context or scenario [8]. On the other hand, trait empathy refers to an individual’s long-term, baseline capacity for empathizing with others. The degree of its malleability remains debated. Some studies suggest that changes in trait empathy are primarily limited to the accumulation of cognitive empathy, as the affective component is known to be influenced by genetic factors [23]. Others contend that repeated training can reliably enhance this ability [1]. Nonetheless, in either case, training in trait empathy requires repeated practice, much like honing a skill or strengthening a muscle [8][23].

2.2 Temporal Misalignment in Empathic VR

A significant challenge arises from the common oversight of a crucial conceptual distinction in practice, leading to methodological mismatches. Enhancing trait empathy through VR necessitates a

longitudinal approach, carefully balancing exposure frequency with representational breadth. Such interventions typically manifest as multi-week training programs, where participants engage with either thematically related [40] or broadly diverse scenarios on a recurring basis [1]. Conversely, the prevalent strategy in empathic VR—replicating a target individual’s or scenario’s lived experience for participant immersion and interaction—is inherently designed to elicit state empathy.

Given their differing temporal dynamics and consolidation mechanisms, it is unsurprising that state and trait empathy necessitate distinct measurement approaches for accurate change capture. Most existing empathy scales are explicitly designed to assess baseline dispositional traits with context-dependent questions [23][9][32]. In contrast, the transient nature of state empathy permits and often necessitates a broader spectrum of measurement approaches. Beyond dedicated scales for state empathy [35], some studies have also explored the use of naturalistic reflection prompts, indirect assessments such as design tasks, and psychophysiological indicators to capture dynamic changes in empathic experience [23][19].

Nonetheless, a substantial portion of current empathic VR studies still conflate these constructs, paradoxically framing their objective as enhancing trait empathy while employing single-session, episodic designs more appropriately suited to eliciting state-level responses. This misalignment between conceptual goals and methodological design seem to result in the use of inappropriate measurement tools, as reflected in the reporting of null results across studies. A common concern is that trait empathy scales lack the sensitivity to detect meaningful changes following a brief VR experience.

Even when researchers explicitly target state empathy, additional challenges arise from the temporal complexity of its subcomponents. Different dimensions of empathy consolidate over different timescales, making measurement highly time-sensitive. This issue is particularly salient in VR, where empathy is shaped not only by immediate affective responses, but also by reflective engagement with the immersive experience.

3 Empathy in VR: A Chronological Progression

In bridging such gaps, the present study outlines the chronological progression of empathy in VR, presenting a temporal model that traces how empathic processes unfold over time and offering corresponding insights into their measurement.

3.1 Event-Related Resonance: Rapid Somatic and Affective Engagement

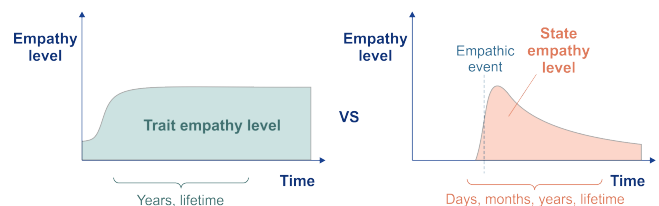


Figure 1: Trait Empathy and State Empathy

Among empathy's components, somatic and affective empathy elicit rapid responses, functioning almost as an event-related resonance within the emotional system immediately following an observed event, as shown in Figure 1. While related, affective and somatic are mediated by different neural mechanisms. Affective empathy is mediated by the inferior frontal gyrus and the insula—structures associated with more primitive, phylogenetically early emotional functions—and thus tends to operate automatically and outside of conscious awareness [4][34]. In contrast, somatic empathy arises from neural-level mirroring processes in response to perceived emotional stimuli and is modulated by activity in the somatosensory cortex [2][30][41]. These mechanisms represent the components of empathy characterized by automatic elicitation and are often discussed in relation to emotional contagion [15].

Correspondingly, both affective and somatic empathy can be assessed using psychophysiological signals such as electroencephalography (EEG), electrodermal activity (EDA), and heart rate variability (HRV). EEG captures brain activity, with mu rhythm suppression (8–13 Hz) serving as an indicator of sensorimotor resonance in the somatosensory cortex [24]. EDA reflects emotional mirroring through increased skin conductance linked to sympathetic arousal [12], and is often interpreted alongside cardiac fluctuations measured by HRV [11]. These methods are particularly effective for detecting affective responses that involve heightened physiological arousal, but are less suited for capturing affective empathy grounded in contextual understanding rather than embodied resonance.

3.2 Reflective Integration: Cognitive Processing and the Formation of Empathic Attitudes

Over a slightly longer timespan, the consolidation of cognitive and compassionate empathy becomes observable, alongside certain forms of affective empathy reinforced by contextual understanding and the projection of social values. These empathic components often involve deliberate mental effort and active perspective-taking [6]. Cognitive empathy engages higher-order neural processes, mediated by regions such as the ventromedial prefrontal cortex, superior temporal sulcus, temporoparietal junction, and medial frontal lobes, and is associated with effortful thinking and projection [10][34]. Compassionate empathy, in contrast, emphasizes the prosocial, behavior-facilitating dimension of empathy and is often regarded as an independent psychological state that is not tied to a specific neural circuit [29].

While changes in cognitive empathy are typically not immediate, they can emerge quickly enough to produce noticeable effects even after a brief 15-minute VR experience [17]. Measuring cognitive empathy usually involves externalizing contextual understanding, taking multiple forms such as scales [35], interviews [13], or problem identification tasks [17]. Compassionate empathy, meanwhile, tends to unfold as the experience deepens—sometimes emerging shortly after the affective response, or taking longer in less familiar or emotionally ambiguous contexts [18]. Although self-report measures can be used, indicators such as willingness to help [14] and prosocial behavior [16] are frequently regarded as more direct reflections of compassionate empathy. Because cognitive empathy is essentially encoded along with knowledge about target scenarios,

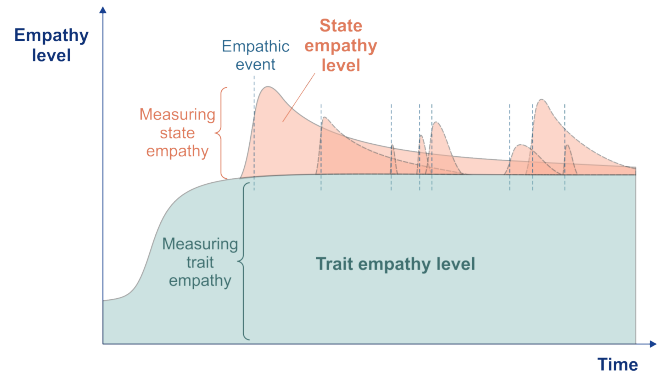


Figure 2: State Empathy Influence Trait Empathy

this type of empathy tends to be more enduring and less prone to rapid decay [23]. In contrast, both compassionate and affective empathy, as transient emotional states, are more susceptible to decline over time. Nonetheless, several studies have shown that helping attitudes and behaviors may persist for weeks or months following the initial VR experience [16][40].

3.3 Dispositional Consolidation: Trait-Level Internalization and Long-Term Impact

It is challenging to pinpoint the exact threshold at which accumulated empathic enhancement leads to observable consolidation in empathy traits. Current literature suggests that four weekly VR experiences over a month can elicit sufficiently strong changes in empathic ability [1]. Notably, this enhancement does not cease with the conclusion of the VR experience; rather, it may continue to develop over subsequent months [40]. Existing models (Figure 2) suggest that empathy traits are longitudinally influenced by empathy states. Specifically, cognitive empathy tends to be retained, serving as a foundation for drawing meaningful parallels between one's own experiences and those of others, thereby enhancing overall empathic ability.

Measuring trait empathy has not posed a major challenge, as the field of psychology has developed a range of validated scales, including the Interpersonal Reactivity Index (IRI), the Questionnaire of Cognitive and Affective Empathy (QCAE), and the Affective and Cognitive Measure of Empathy (ACME). Among these, the IRI has been the most widely used [23], although recent findings suggest that the construct validity of the ACME may surpass that of the IRI. The major pitfall, however, lies in the misapplication of these trait-based scales in single-session studies, where they may fail to capture transient, state-level changes in empathy.

4 Empathic VR Design Through a Temporal Lens

The temporal progression of empathy's diverse components dictates differential design considerations and recommendations for designers for optimally facilitating each stage within VR interventions.

4.1 Design for Event-Relevant Resonance

Empathic VR scenarios designed to elicit rapid, event-related somatic and affective responses typically emphasize immersive sensory input to provoke immediate emotional resonance occurring within hundreds of milliseconds after the onset of the event. This approach is frequently adopted in immersive journalism, where empathy is elicited through direct exposure to the suffering, pain, or loss experienced by individuals and communities affected by humanitarian crises [5][22]. Because this design strategy focuses on perceptual immediacy and emotional intensity, it tends to engage cognitive empathy to a lesser extent [28]. In many respects, designing for such immediate empathy parallels the creation of immersive film, a similarity that reflects the medium's origins in cinematic storytelling [5].

For this type of VR, the sense of presence and immersion serves as the foundation for emotional engagement, while narrative scaffolds the experience and guides interpretation. Interactivity is often minimal or absent, as the primary goal is not to shape the environment but to witness and internalize emotionally charged experiences. Accordingly, for interventions targeting this form of empathy, the functional differences among fully immersive VR, 360-degree video (VR-360), and simpler formats such as Cardboard VR are frequently regarded as marginal or even negligible [31].

4.2 Design for Reflection Integration

By contrast, VR scenarios designed to foster reflective empathy rely more heavily on interactivity and are structurally closer to role-playing games than films. In such experiences, participants actively engage with complex, context-rich situations that support the development and consolidation of empathic understanding over time. This design strategy is more common in domains such as medical training, education, historical simulation, and empathic co-design, where the goal is to facilitate perspective-taking and deeper cognitive and social insight [36][26]. While these scenarios often elicit affective responses, cognitive empathy is typically more pronounced and enduring in its impact [23]. As mentioned in the previous section, this form of empathy began to emerge after approximately 15 minutes of VR exposure and continued to consolidate over the following weeks.

Interactive tasks are central to this form of empathic VR, as they expose users to nuanced dilemmas and conflicting priorities that mirror real-world complexity. As a result, the differences between fully immersive VR and VR-360 become especially salient, with design strategies for one format proving largely non-transferable to the other [38]. Among key design considerations, logical validity—the extent to which the environment behaves in ways that align with real-world reasoning and user expectations—has been identified as particularly critical. Breakdowns in this logic can quickly erode presence, reminding participants of the simulation's artificial nature. In contrast, visual fidelity, while appreciated, is generally considered less central; studies have demonstrated that visually simplified environments can still evoke strong empathic responses [20].

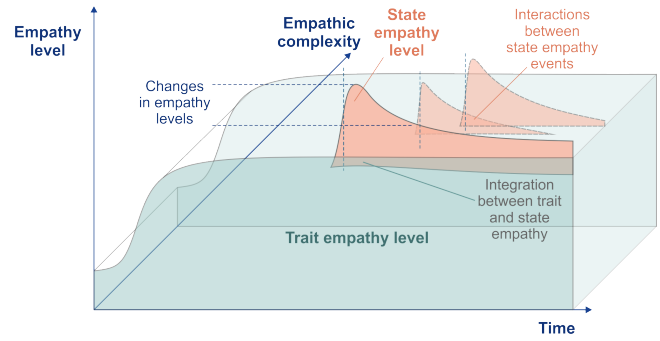


Figure 3: Design for Temporal Progression of Empathy

4.3 Design for Trait Empathy

There are relatively few examples of VR experiences explicitly designed to enhance trait empathy, in part because the distinction between trait and state empathy remains under-recognized in much of the literature. Among the existing cases, design strategies vary considerably. For instance, Barbot and Kaufman [1] implemented a five-week program in which participants engaged with 25 distinct VR scenarios. Rather than focusing on conventionally empathic content, these scenarios emphasized creativity through perspective-taking, environmental exploration, storytelling, and problem-solving. Their findings suggest that experiential dimensions—such as immersion, presence, and agency—played a more decisive role in fostering empathic outcomes than the thematic content itself, as indicated by their factorial analysis.

In contrast, Wijma et al. [40] developed a medical training program where caregivers repeatedly engaged with VR scenarios simulating patients' lived challenges. This long-term exposure led to significant increases in both affective and cognitive empathy toward patients. Participants' reflections in this study indicated that the theme and task were critical to their experience, with the knowledge and perspective gained proving extremely helpful.

Given the divergent nature of these two approaches, drawing definitive conclusions about which specific design dimensions are most critical is challenging. However, one consistent finding emerges: repeated exposure over time appears to be a necessary condition for facilitating meaningful changes in trait empathy. Moreover, when the VR experience engages with aspects of empathy, the cumulative changes in participants' empathic states are posited to influence their long-term empathic capacity—potentially lasting for months, as illustrated in Figure 3.

5 Conclusion

This study has addressed a critical yet underexplored dimension in empathic VR research: the temporal dynamics of empathy and their implications for effective VR design. By examining how different components of empathy unfold across time—from rapid, automatic somatic and affective responses to slower, cognitively mediated and potentially enduring forms such as compassionate and trait empathy—it highlights the importance of aligning design strategies and measurement approaches with the specific temporal profile of the empathic response being targeted. The analysis demonstrates

that empathy is not a unitary construct but a temporally structured process, each phase of which calls for distinct forms of engagement and evaluation. Designing effective empathic VR therefore requires a nuanced understanding of how these components evolve, consolidate, and potentially contribute to long-term empathic change.

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